



Pushparaj E



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PROFESSIONAL SUMMARY

Results-driven Software Engineer with **3 years of experience in full-stack development, enterprise ERP systems, and AI-powered automation**. Proven success building scalable CRM and ERP platforms that streamline operations for **65+ internal users**. Specialized in **JavaScript, Node.js, React, and PostgreSQL**, with hands-on expertise in predictive maintenance systems and document automation using LLM technologies. Built and scaled CRM systems handling **4,900+ leads**, transforming manual workflows into data-driven software solutions that **reduce operational overhead by 40%** and improve cross-functional efficiency.



TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Python, SQL, HTML5, CSS3

Frameworks & Libraries: Node.js, Express.js, React.js, LangChain, Next.js

Databases: PostgreSQL, MongoDB, Firebase Realtime Database

DevOps & Tools: Git, Postman, Jira, Figma, CI/CD, Render, Linode

AI/ML Technologies: LLM Integration, Ollama, Groq, AI Agents, Predictive Analytics, Anomaly Detection



PROFESSIONAL EXPERIENCE

Petrokens Engineering and Services Pvt Ltd

07/2025 | Tharamani, Chennai

ProjectEPC – Enterprise ERP Platform (Production)

Tech: React.js, Node.js, Express.js, PostgreSQL, REST APIs, JWT, RBAC, Tailwind CSS

Overview: Enterprise ERP system centralizing operations, automating workflows, and enabling real-time reporting for **65+ users**, reducing bottlenecks by **45%**.

Roles & Responsibilities:

- Developed core ERP modules: **CRM, Project Management, Timesheets, HRMS, Role-based Dashboards, and Proposal System**
- Architected and deployed a company-wide ERP platform for **65+ users**, reducing operational bottlenecks by **45%**
- Designed scalable backend with **130+ PostgreSQL tables** and **500+ REST APIs**, enabling real-time cross-department data flow
- Optimized database performance, eliminating **60% data redundancy** and improving reporting speed by **3x**
- Implemented secure **RBAC with audit logging**, achieving zero unauthorized access incidents
- Automated document workflows and approval processes, reducing manual inter-department coordination

2. AI-Powered Pump Predictive Maintenance System (In Development)

- Developing an ML-based predictive maintenance system analyzing **vibration, temperature, and pressure** data
- Processing **10,000+ daily sensor readings** using anomaly detection and time-series models
- Performing feature engineering and data preprocessing for fault pattern identification
- Building alert mechanisms to support proactive maintenance and reduce unplanned downtime

3. AI Document Intelligence for QA/QC (R&D Prototype)

- Building LLM-powered system to analyze and summarize **100+ page engineering documents**
- Implementing chunking, embeddings, and vector retrieval for semantic document search

- Designing contextual memory pipelines for accurate long-document understanding
- Developing AI workflows for automated QA/QC document review and insight extraction

2. Urbancode Edutech Solutions PVT LTD

04/2024 – 07/2025
Pallikaranai, Chennai

Zen CRM – Training & Operations Management Platform

Tech: React.js, Node.js, Express.js, PostgreSQL, Tailwind CSS, Material UI, Axios, JWT, Linode

Overview: Enterprise CRM system for managing student lifecycle, training operations, payments, and internal workflows with RBAC, automation, and analytics.

Roles & Responsibilities:

- Architected an enterprise-grade CRM platform managing the complete student lifecycle and processing **4,900+ leads**
- Developed **70+ secure REST APIs** using Node.js and Express.js with JWT authentication and Role-Based Access Control (RBAC), achieving response times under **200ms**
- Designed a structured **12-stage workflow system** (Enquiry, Enrolled, On-Hold, Archived, etc.) with real-time dashboards and status tracking
- Built multi-step lead forms, dynamic filtering features, and a **Quill.js-based internal commenting system**
- Integrated automated communication workflows including brochure sharing, follow-ups, and class reminders using Google Mail API and scheduled scripts
- Developed modules for course, batch, and trainer management with full CRUD functionality and relational data linking
- Implemented operational features such as bulk lead uploads, fee reminders, and **Payment Insights dashboards** for financial tracking
- Managed end-to-end system delivery including responsive frontend development with Tailwind CSS, backend architecture, PostgreSQL schema design, and **Linode** cloud deployment

3. Leather Sector Skill Council (LSSC)

06/2023 – 05/2024
Nungambakkam, Chennai

Scale LMS – Training platform for vocational upskilling

Overview: LMS platform for the leather sector enabling vocational training, real-time progress tracking, skill assessment, and performance analytics for trainers and learners.

Tech: Node.js, Express.js, Firebase, HTML, CSS, JavaScript, Bootstrap

Roles & Responsibilities:

- Developed modules for real-time progress tracking, interactive assessments, and role-based access(RBAC)
- Built analytics dashboards for trainers to monitor learner performance
- Designed responsive UI using Bootstrap and implemented backend logic in Express.js
- Integrated Firebase for secure data handling and real-time data sync
- Collaborated across teams to improve training visibility and learner engagement

 ACHIEVEMENTS

- Trained **50+ students** in MERN stack, Python, and Java through mentoring and technical training
- Built **3+ production LMS and CRM platforms** for real-world organizations
- Delivered **5+ client websites**, improving their digital presence and online reach
- Developed enterprise ERP modules supporting operations for **65+ users**
- Contributed to AI-based automation systems for predictive maintenance and document intelligence

 EDUCATION

B.E.Computer Science Engineering, Kings Engineering College | CGPA - 8.5

2021-2025

 SOFT SKILLS

Team Collaboration | Problem Solving | Time Management | Mentoring & Leadership | Communication